

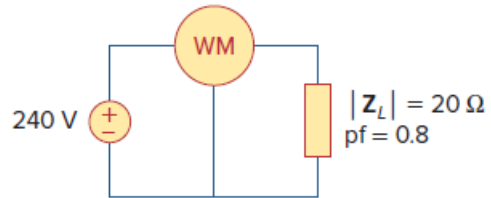
Problem

The circuit of Fig. 11.95 portrays a wattmeter connected into an ac network.

(a) Find the magnitude of the load current.

(b) Calculate the wattmeter reading.

Figure 11.95:



Step-by-step solution

Step 1 of 4

Refer to Figure 11.95 in the text book.

Convert the value of the voltage to RMS value and represent in phasor form.

$$\begin{aligned} V_s &= \frac{240}{\sqrt{2}} \angle 0^\circ \\ &= 169.71 \angle 0^\circ \text{ V rms} \end{aligned}$$

Therefore, the value of the source voltage in phasor form is, $169.71 \angle 0^\circ \text{ V rms}$.

Step 2 of 4

Calculate the value of the load.

$$Z_L = |Z_L| \angle \cos^{-1}(\text{pf})$$

Substitute 20Ω for $|Z_L|$ and 0.8 for pf .

$$\begin{aligned} Z_L &= 20 \angle \cos^{-1}(0.8) \\ &= 20 \angle 36.87^\circ \\ &= 16 + j12 \Omega \end{aligned}$$

Therefore, the value of the load impedance Z_L is $16 + j12 \Omega$.

Step 3 of 4

(a)

Calculate the value of the current I_{rms} passing in the circuit.

$$I_{\text{rms}} = \frac{V_s}{Z_L}$$

Substitute $16 + j12 \, \Omega$ for Z_L and $169.71 \angle 0^\circ \text{ V rms}$ for V_s .

$$\begin{aligned} I_{\text{rms}} &= \frac{169.71 \angle 0^\circ}{16 + j12} \\ &= \frac{169.71 \angle 0^\circ}{20 \angle 36.87^\circ} \\ &= 8.49 \angle -36.87^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_{rms} is $8.49 \angle -36.87^\circ \text{ A rms}$.

Therefore, the magnitude of the load impedance is 8.49 A rms.

Step 4 of 4

(b)

Notice that Wattmeter reads only the average power absorbed by the load.

Calculate the value of the average power absorbed by the load.

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Substitute 169.71 V rms for V_{rms} , 0° for θ_v , -36.87° for θ_i and 8.49 A rms for I_{rms} .

$$\begin{aligned} P &= (169.71)(8.49) \cos(0^\circ - (-36.87^\circ)) \\ &= (1440.84) \cos(0^\circ + 36.87^\circ) \\ &= (1440.84) \cos(36.87^\circ) \\ &= (1440.84)(0.8) \end{aligned}$$

$$P = 1152.67 \text{ W}$$

Therefore, the value of the wattmeter reading is, 1152.67 W.